G12Adv	Mathematics	الرياضيات	T1 - 2019/20

Multiple Choice Part	جزء الاختيار من متعدد
Show all your work when answering these questions.	يجب كتابة خطوات الحل التفصيلية للمفردات الاختبارية كافة.

Multiple Choice — اختيار من متعدد

1. Find $\lim_{x\to 2}\left(\frac{1}{x-2}-\frac{4}{x^2-4}\right)$ if it exists.

A) 2

B) $\frac{1}{2}$

C) $\frac{1}{4}$

D) Does not exist

2. Find $\lim_{x\to 0}\left(\frac{1-\cos^2 x}{x^2}\right)$ if it exists.

A) 0

B) 1

C) 2

D) Does not exist

3. Which of the following functions is not differentiable at $x=2$?

A) $f(x)=\begin{cases}4, & x<2\\ 2x, & x\geq 2\end{cases}$

B) $g(x)=\frac{2}{x+2}$

C) $h(x)=|x-2|^2$

D) $Q(x)=x^2-4$

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4. **Find the derivative of the function** $f(x) = \tan^{-1}(\cos 2x)$.

A) $f'(x) = \dfrac{2}{1+4x^2}$

B) $f'(x) = \dfrac{-2\sin 2x}{1+(\cos 2x)^2}$

C) $f'(x) = \dfrac{2}{\sqrt{1-(\cos 2x)^2}}$

D) $f'(x) = \dfrac{2\sin 2x}{1+(\cos 2x)^2}$

5. **Find** $g'(1)$ **if** $g(x) = \dfrac{4x^2 - x + 3}{\sqrt{x}}$.

A) $g'(1) = 6$

B) $g'(1) = 4$

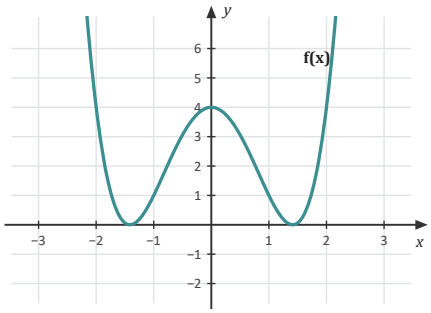
C) $g'(1) = 10$

D) $g'(1) = 5$

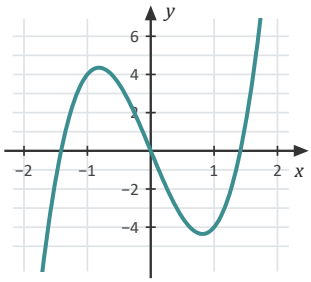
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Multiple Choice	اختيار من متعدد

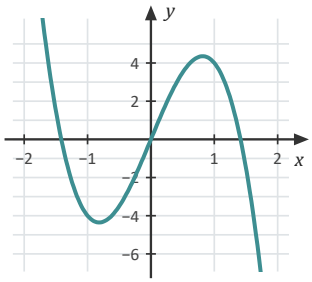
6. Use the graph of $y = f(x)$ to sketch a graph of $y = f'(x)$.



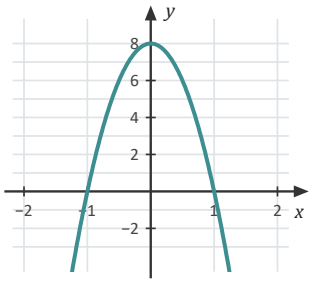
A)



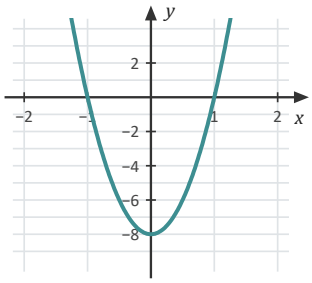
B)



C)



D)



7. Find the derivative of the function $f(t) = \ln(t^3 + 3t)$.

A) $f'(t) = (3t^2 + 3)\ln(t^3 + 3t)$

B) $f'(t) = \dfrac{1}{t^3 + 3t}$

C) $f'(t) = \dfrac{3t^2 + 3}{t^3 + 3t}$

D) $f'(t) = \dfrac{3t^2 + 3}{(t^3 + 3t)^2}$

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8. **Find** $\lim_{x\to 0}\left(\frac{2x-|x|}{|3x|-2x}\right)$ **if it exists.**

A) 0

B) 1

C) $-\frac{3}{5}$

D) Does not exist

9. **Find all the points at which the function** $f(x)$ **is discontinuous, and determine which of them is removable.**

$$f(x)=\begin{cases}\sin x, & x<0\\ x^2, & 0\le x\le 2\\ 4x-3, & x>2\end{cases}$$

A) f has a non-removable discontinuity at x = 2

B) f has a removable discontinuity at x = 2

C) f has a non-removable discontinuity at x = 0

D) f has a removable discontinuity at x = 0

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10. **The concentration of a certain chemical after t seconds of an autocatalytic reaction is given by $c(t) = \frac{6}{2e^{-8t}+1}$. (a) Show that $c'(t) > 0$. (b) Use this information to determine that the concentration of the chemical never exceeds 6.**

A) c'(t) > 0 and the concentration of the chemical never exceeds 6

B) c'(t) < 0 and the concentration of the chemical never exceeds 6

C) c'(t) > 0 and the concentration of the chemical exceeds 6

D) c'(t) > 0 and the concentration of the chemical never exceeds 3

11. **Use logarithmic differentiation to find the derivative of $f(x) = (x^2)^{3x}, x > 0$.**

A) $g'(x) = 6(\ln x + 2)(x^2)^{3x}$

B) $g'(x) = 6(\ln x + 1)(x^2)^{3x}$

C) $g'(x) = 3(\ln x + 1)(x^2)^{3x}$

D) $g'(x) = 2(\ln x + 1)(x^2)^{3x}$

12. **Suppose that the current price is AED 2.40 and 12,000 items are sold at that price. If the price is increasing at the rate of 10 fils per year and the quantity sold decreases at the rate of 1500 items per year, at what rate is the revenue changing? (Hint: Revenue equals price times quantity.)**

A) Revenue is increasing at a rate of AED 2,400 per year

B) Revenue is decreasing at a rate of AED 1,500 per year

C) Revenue is increasing at a rate of AED 1,500 per year

D) Revenue is decreasing at a rate of AED 2,400 per year

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13. If $f(x) = 2x^4 + bx + 3$ and $\lim_{h\to 0} \dfrac{f(2+h) - f(2)}{h} = 10$, find the value of b.

A) $b = -54$ **B)** $b = 54$

C) $b = -10$ **D)** $b = 10$

14. Find $y'(x)$ implicitly for $\dfrac{x}{y} + \dfrac{2}{x} = 5$.

A) $y' = \dfrac{2x - 5y}{5x - 2}$ or $y' = \dfrac{x^2y - 2y^2}{x^3}$

B) $y' = \dfrac{5x - 2y}{5x - 2}$ or $y' = \dfrac{x^2y^2 + 2y}{x^3}$

C) $y' = \dfrac{-2x + 5y}{-5x - 2}$ or $y' = \dfrac{x^2y + 2y^2}{2x^3}$

D) $y' = \dfrac{x - y}{5x + 2}$ or $y' = \dfrac{2x^2y + y^2}{3x^3}$

15. Find the value of c satisfying the conclusion of the Mean Value Theorem for $f(x) = x^2 + 2$ on $[0, 2]$.

A) $c = 1$ **B)** $c = 2$

C) $c = 0.5$ **D)** $c = 0$

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16. **Find the derivative of** $f(x) = \left(\sqrt{x^2+4} - 3x^2\right)^{3/2}$.

A) $f'(x) = \frac{3}{2}\left(\sqrt{x^2+4} - 3x^2\right)^{1/2}\left(\frac{x}{\sqrt{x^2+4}} - 6x\right)$

B) $f'(x) = \left(\sqrt{x^2+4} - 3x^2\right)^{1/2}\left(\frac{x}{\sqrt{x^2+4}} - 6x\right)$

C) $f'(x) = \frac{3}{2}\left(\sqrt{x^2+4} - 3x^2\right)^{1/2}\left(\frac{2x}{\sqrt{x^2+4}} - 6x\right)$

D) $f'(x) = \frac{3}{2}\left(\sqrt{x^2+4} - 3x^2\right)^{1/2}\left(\frac{x}{2\sqrt{x^2+4}} - 6x\right)$

17. **Find the equation of the tangent line to** $y = f(x) = x\sin x$ **at** $x = \frac{\pi}{2}$.

A) $y = -x$ **B)** $y = x$

C) $y = 2x$ **D)** $y = 3x$

18. **If a value of a 100-dirham investment doubles every year, its value after** t **years is given by** $v(t) = 100(2)^t$. **Find the instantaneous percentage rate of change of the value.**

A) 69.3% per year **B)** 39.3% per year

C) 59.3% per year **D)** 49.3% per year

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19. **Find the derivative of** $y = \ln \sqrt{\sin 4x}$.

A) $\frac{dy}{dx} = -2 \cot 4x$

B) $\frac{dy}{dx} = 2 \cot 4x$

C) $\frac{dy}{dx} = \cot 4x$

D) $\frac{dy}{dx} = 2 \tan 4x$

20. **Find all value(s) of** x **for which the tangent line to** $y = xe^{-2x}$ **is horizontal.**

A) $x = -\frac{1}{4}$

B) $x = \frac{1}{2}$

C) $x = \frac{1}{4}$

D) $x = -\frac{1}{2}$